

Recollection G group, $\text{Aut}(G) := \{\text{automorphisms } G \simeq G\}$, $\text{Ad}_g := x \mapsto gxg^{-1} \in \text{Aut}(G)$.

Note that $\text{Ad}_g = \text{id}_G \iff gxg^{-1} = x, \forall x \iff gx = xg \iff g \in Z_G$: center of G , subgroup.

§ Cyclic groups

Definition G group. We say G is cyclic if $\exists \sigma \in G, G = \langle \sigma \rangle$, i.e. $\forall g \in G, \exists k \in \mathbb{Z}, g = \sigma^k$.

Example $\forall n \in \mathbb{Z}, (\mathbb{Z}/n\mathbb{Z}, +)$ is cyclic, $\therefore$ can take $\sigma = 1 + n\mathbb{Z}$.

Want to classify all cyclic groups, up to isomorphism.

Proposition Let $G = \langle \sigma \rangle$ be cyclic.

- If $|G| = \infty$ then $\mathbb{Z} \simeq G$ via $n \mapsto \sigma^n$.
- If $n := |G| < \infty$, then $\mathbb{Z}/n\mathbb{Z} \simeq G$ via $k + n\mathbb{Z} \mapsto \sigma^k$.

Proof Can define the homomorphism (surjective) $\mathbb{Z} \twoheadrightarrow G, k \mapsto \sigma^k$.

- If it is injective, we get $\simeq, |G| = |\mathbb{Z}| = \infty$.
- If not injective, namely $i < j$ s.t. $\sigma^i = \sigma^j \implies \sigma^{j-i} = 1$. Can define $n := \min\{k \geq 1 \mid \sigma^k = 1\}$.

$\forall k \in \mathbb{Z}, \sigma^k$ depends only on $k \bmod n, \therefore \sigma^n = 1$.

$\implies$ Get $\mathbb{Z}/n\mathbb{Z} \twoheadrightarrow G, k + n\mathbb{Z} \mapsto \sigma^k$ still a homomorphism.

Injectivity: suppose $\sigma^i = \sigma^j, 0 \leq i \leq j < n$. Then $\sigma^{j-i} = 1, 0 \leq j-i < n$.

$\implies$ must have $i = j$ by the minimality of n .

Hence $\mathbb{Z}/n\mathbb{Z} \simeq G, |G| = n$. □

As conclusion, cyclic groups are (up to $\simeq$) $\begin{cases} \mathbb{Z} & (\text{infinite}) \\ \mathbb{Z}/n\mathbb{Z}, n \geq 1 & (\text{finite}) \end{cases}$

$n \neq m \implies \mathbb{Z}/n\mathbb{Z} \not\simeq \mathbb{Z}/m\mathbb{Z}$ by counting elements.

§ Cosets in a group

Let G group, $H \subset G$ subgroup (sometimes denoted as $\leq$).

$\forall x, y \in G$, define $x \underset{\text{left}}{\sim} y$ if $\exists h \in H$ s.t. $x = hy$. The right version is similar.

This is a equivalence relation on G :

- $x \underset{\text{left}}{\sim} x, \therefore x = 1_H \cdot x$,
- $\left(x \underset{\text{left}}{\sim} y \implies y \underset{\text{left}}{\sim} x \right), \therefore x = hy \implies y = h^{-1}x$,
- $\left(x \underset{\text{left}}{\sim} y, y \underset{\text{left}}{\sim} z \implies x \underset{\text{left}}{\sim} z \right), \therefore x = hy, y = h'z \implies x = (hh')z$.

Recall X set, $\sim$ equivalence relation on X , then

$$X/\sim := \{\sim \text{ equivalence classes in } X\}$$

where $X = \bigsqcup_{C \in X/\sim} C$, with quotient map $X \rightarrow X/\sim$.

Now: $\forall g \in G$, set $Hg := \{hg \mid h \in H\}, gH := \{gh \mid h \in H\}$. They are the left (resp. right) equivalence class of g , and called the right (resp. left) coset of g .

The corresponding quotient sets are

$$H \backslash G := G/\underset{\text{left}}{\sim} = \{\text{right cosets } Hg \mid g \in G\},$$

$$G/H := G/\underset{\text{right}}{\sim} = \{\text{left cosets } gH \mid g \in G\}.$$

$$\text{So } G = \bigsqcup_{Hg \in H \backslash G} Hg = \bigsqcup_{gH \in G/H} gH.$$

Lemma $\forall$ subset $S \subset G, S^{-1} := \{s^{-1} \mid s \in S\} \subset G$
 $\implies (Hg)^{-1} = g^{-1}H^{-1} = g^{-1}H, (gH)^{-1} = Hg^{-1}.$

Proposition $_H \backslash G \leftrightarrow G/_H$ via

$$Hg \mapsto g^{-1}H = (Hg)^{-1},$$

$$(gH)^{-1} = Hg^{-1} \mapsto gH.$$

Proof Both directions are given by $S \mapsto S^{-1}$, and $(S^{-1})^{-1} = S.$ □

Definition $(G : H) :=$ the common cardinality of $_H \backslash G, G/_H$, called the **index** of $H \subset G$.

Theorem (Lagrange) $|G| = |G| \cdot (G : H).$ (multiplication of cardinal numbers)

Proof $\forall$ coset $C \in _H \backslash G$, pick $x_c \in C$, i.e. $Hx_c = C$. Consider

$$H \times _H \backslash G \rightarrow G,$$

$$(h, C) \mapsto hx_c,$$

Surjective: $\forall g \in G, \exists C$ s.t. $g \in C = Hx_c \implies g \in \text{im}.$

Injective: $hx_c = h'x_{c'} \implies C \cap C' \neq \emptyset \implies C = C', x_c = x_{c'}$. Cancellation law gives $h = h'.$

Hence $|G| = |H| \cdot |_H \backslash G| = |H| \cdot (G : H).$

Another way: $G = \bigsqcup_{C \in _H \backslash G} C$; for each C we have $|C| = |H|$ since cancellation law $\implies Hg \leftrightarrow H$.

This also gives $|G| = |H| \cdot |_H \backslash G|.$ □

Corollary G finite, then $|H|$ divides $|G|$ for all subgroups $H \subset G$.

Corollary $H, K \subset G$ subgroups, finite. $|H|, |K|$ coprime $\implies H \cap K = \{1\}.$

Proof $H \cap K$ is a subgroup of both H and K , hence $|H \cap K|$ divides $|H|, |K| \implies |H \cap K| = 1.$
 $\implies H \cap K = \{1\}.$ □

Definition G group, $\sigma \in G$. **order** / ord of σ : $\text{ord}(\sigma) = |\langle \sigma \rangle|.$

Recall that $\text{ord}(\sigma) = \min\{k \geq 1 \mid \sigma^k = 1\}.$

Moreover:

$$\sigma^k = \sigma^l \iff \text{ord}(\sigma) \mid |k - l|,$$

$$\sigma^k = 1 \iff \text{ord}(\sigma) \mid k.$$

i.e. $\text{ord}(\sigma) =$ “the minimal period of $\sigma^0, \sigma^1, \sigma^2 \dots$ ”

Corollary G finite, $\sigma \in G$, then $\text{ord}(\sigma) \mid |G|.$

Corollary If $p := |G|$ is a prime number, then G is cyclic.

Proof Note that G : any group, $\sigma \in G, \text{ord}(\sigma) = 1 \iff \sigma = 1.$

Pick $\sigma \in G, \sigma \neq 1.$

$$1 \neq \text{ord}(\sigma) \mid p \implies \text{ord}(\sigma) = p \implies |\langle \sigma \rangle| = |G|.$$

Another property of indices: $K \subset H \subset G$ subgroups, then $(G : K) = (G : H)(H : K).$

Proof $G = \bigsqcup_{C \in H \setminus G} Hx_c$, pick $x_c \in C$ for all C ; $H = \bigsqcup_{D \in K \setminus H} Ky_d$, pick $y_d \in D$ for all D .

$$\begin{aligned} G &= \bigsqcup_C Hx_c = \bigsqcup_{C,D} Ky_dx_c \\ \implies {}_K \setminus G &\longleftrightarrow \{(C,D) \mid C \in {}_H \setminus G, D \in {}_K \setminus H\} \\ \implies (G : K) &= (G : H)(H : K). \end{aligned}$$

□

§ Groups in action

In the earlier examples, groups “act” on a set or other structures.

Let X : non-empty set, $S_x := \{\text{permutations on } X\}$ acts on X , and S_n acts on $\{1, 2, \dots, n\}$.

$\forall F$ -vs. V , $\text{GL}(V) = \{\text{automorphisms } V \xrightarrow{\sim} V\}$ acts on V .

G group, $\text{Aut}(G)$ acts on G .

Definition G group, X set. A **left G -action on X** means a map $a : G \times X \rightarrow X$ s.t.

- $a(1, x) = x$,
- $a(g, a(h, x)) = a(gh, x)$.

Write $a(g, x) = "gx"$. The properties above becomes $1 \cdot x = x, g(hx) = (gh)x$.

Right G -action on X means $a' : X \times G \rightarrow X$ s.t. $x \cdot 1 = x, (xg)h = x(gh)$ using similar notations.

For convenience, by default action = left action.

Definition G : any group, define its **opposite group** $G^{\text{op}} := (G, \odot)$ s.t. $g \odot g' = g'g$.
 $\implies G^{\text{op}}$ is a group, in which $g^{-1}, 1$ are the same as those in G .

$(G^{\text{op}})^{\text{op}} = G$, and $G^{\text{op}} = G \iff G$ is abelian.

Moreover $G \xrightarrow{\sim} G^{\text{op}}$ via $g \mapsto g^{-1}$.

Left G -action on X = Right G^{op} -action on X . Given $a : G \times X \rightarrow X$, can define $a' : X \times G \rightarrow X$,
 $a'(x, g) = a(g, x)$.

Notation $G \curvearrowright X$ or $X \curvearrowleft G$.

Definition Let $G \curvearrowright X, x \in X$.

- The **G -orbit** of $x = Gx := \{gx \mid g \in G\} \subset X$.
- **Stabilizer** of $x = \text{Stab}_G(x) := \{g \in G \mid gx = x\} \subset G$ subgroup.

$\forall x, y \in X$, write $x \underset{G}{\sim} y$ if $\exists g \in G, x = yg$. This is a equivalence relation on X .

- $\underset{G}{\sim}$ equivalence classes = G -orbits Gx in X .

$${}_G \setminus X := X / \underset{G}{\sim} = \{\text{all } G\text{-orbits } Gx\}.$$

Next time: Decomposition into orbits.